Reasoning test

For admission to Class 12

About the test

- 30 questions in 30 minutes: 10 each of maths, English and science, mixed together.
- Half the maths and science questions are from the Class 11 syllabus (the chapters listed below); the rest test reasoning. Each question is marked Basics or Class 11.
- Taken online on a phone or computer, through a link sent to the parent's WhatsApp. The link works for 2 hours.
- Most questions have four options; a few ask for a number to be typed in.
- One mark for a correct answer. A quarter mark off for a wrong multiple-choice answer; nothing off for a wrong type-in answer or a blank.
- Answers can be changed before submitting. If the candidate leaves the page, the timer pauses.

What the test covers: basics

Basics questions test reasoning, not memory, using ideas taught up to Class 11. Chapters are from the NCERT (CBSE) books a student joining Class 12 in 2027 will have used; other boards teach the same ideas under their own names.

Maths

Class 6 · Mathematics

- *Whole Numbers* – number patterns and series
- *Playing with Numbers* – primes, factors, HCF and LCM
- *Fractions*
- *Mensuration* – perimeter and area
- *Ratio and Proportion*

Class 7 · Mathematics

- *Fractions and Decimals* – fractions
- *Data Handling* – averages (mean)
- *Simple Equations*
- *Comparing Quantities* – percentages, profit and loss, discounts, interest; ratio and proportion
- *Perimeter and Area*

Class 7 · Science

- *Motion and Time* – speed, distance and time

Class 8 · Mathematics

- *Linear Equations in One Variable* – simple equations
- *Squares and Square Roots* – squares and cubes
- *Cubes and Cube Roots* – squares and cubes
- *Comparing Quantities* – percentages, profit and loss, discounts, interest

Class 10 · Mathematics

- *Arithmetic Progressions* – number patterns and series
- *Probability*

Reasoning, not a textbook chapter: Clocks and calendars · Number codes and puzzles · Everyday units · Multi-step word problems · Time and work, pipes and tanks · Problems on ages · Mixtures · Overlapping groups (Venn diagrams)

English

Grammar, vocabulary and reasoning; nothing from the English reader: Vocabulary: meaning in context, opposites, idioms · Grammar: agreement, prepositions, tenses, linking words · Completing and ordering sentences · Analogies, odd one out, letter series and codes · Reading: inference, tone, assumptions, judging an argument · Logical deduction: statements, arrangements, family relations

Science

Class 6 · Science

- *Fun with Magnets* – magnets
- *Light, Shadows and Reflections* – light, shadows and plane mirrors
- *Electricity and Circuits* – electric circuits

Class 7 · Science

- *Heat* – heat and conductors
- *Acids, Bases and Salts* – acids, bases and indicators
- *Physical and Chemical Changes* – physical and chemical changes, rusting
- *Nutrition in Plants* – photosynthesis and plant respiration
- *Respiration in Organisms* – respiration; photosynthesis and plant respiration
- *Motion and Time* – speed
- *Light* – light, shadows and plane mirrors

Class 8 · Science

- *Force and Pressure* – force and pressure, pressure in liquids
- *Sound* – sound, and the speeds of light and sound

Class 9 · Science

- *Matter in Our Surroundings* – evaporation and condensation; change of state
- *Is Matter Around Us Pure?* – elements, compounds, mixtures and solutions
- *Force and Laws of Motion* – inertia, action and reaction
- *Gravitation* – mass and weight; floating, sinking and density
- *Work and Energy* – energy changes

Class 10 · Science

- *Electricity* – electric power and safety
- *Our Environment* – food chains

Reasoning, not a textbook chapter: Reading an experiment and fair tests · Applying a principle to a new situation

Class 11 syllabus questions

Five maths and five science questions come from these Class 11 chapters only: the first-term chapters, because the test can be taken before the year ends.

Maths

Class 11 · Mathematics

- *Ch 1. Sets* – operations on sets, subsets
- *Ch 2. Relations and Functions* – Cartesian products, functions
- *Ch 3. Trigonometric Functions* – radian measure, trigonometric values
- *Ch 4. Complex Numbers and Quadratic Equations* – powers of i, modulus, complex roots
- *Ch 5. Linear Inequalities* – solving inequalities
- *Ch 6. Permutations and Combinations* – arrangements and selections
- *Ch 7. Binomial Theorem* – binomial expansion
- *Ch 8. Sequences and Series* – geometric progressions

Science

Class 11 · Physics

- *Ch 1. Units and Measurement* – SI units, dimensions
- *Ch 2. Motion in a Straight Line* – free fall, equations of motion
- *Ch 3. Motion in a Plane* – vectors, projectiles
- *Ch 4. Laws of Motion* – Newton's laws, impulse, momentum
- *Ch 5. Work, Energy and Power* – work, energy, power

Class 11 · Chemistry

- *Ch 1. Some Basic Concepts of Chemistry* – the mole, stoichiometry
- *Ch 2. Structure of Atom* – protons, neutrons, electron configuration
- *Ch 3. Classification of Elements and Periodicity in Properties* – periodic trends
- *Ch 4. Chemical Bonding and Molecular Structure* – bonds, shapes of molecules
- *Ch 7. Redox Reactions* – oxidation numbers

Sample paper

These questions were written for this sheet. The real test has different questions, in the same style and at a similar level.

1. MATHS · BASICS
   Find the next number in the series: 1, 2, 6, 24, 120, ___

   A 240 **B** 600
   C 720 **D** 840

2. ENGLISH · BASICS
   Choose the word most nearly opposite in meaning to "transparent".

   A opaque **B** clear
   C bright **D** thin

3. SCIENCE · BASICS
   When a hanging carpet is beaten with a stick, dust falls out of it. This is mainly because

   A the dust stays put as the carpet moves **B** the stick attracts the dust towards it
   C the dust becomes heavier when it is hit **D** the air pushes the dust inwards

4. MATHS · CLASS 11
   If A = {2, 4, 6} and B = {4, 6, 8, 10}, how many elements are there in A ∪ B?

   Type-in answer: _____

5. ENGLISH · BASICS
   The soldier was praised for his ___ in rescuing the trapped villagers.

   A cowardice **B** courage
   C carelessness **D** confusion

6. SCIENCE · CLASS 11
   A car speeds up uniformly from rest to 20 m/s in 4 seconds. How far does it travel in that time, in metres?

   Type-in answer: _____

7. MATHS · BASICS
   A bicycle bought for ₹4,000 is sold at a loss of 15%. What is the selling price, in rupees?

   Type-in answer: _____

8. ENGLISH · BASICS
   Choose the correct words: By the time we reached the station, the train ___.

   A has left **B** leaves
   C is leaving **D** had left

9. SCIENCE · BASICS
   Why is it dangerous to operate an electric switch with wet hands?

   A Wet hands slip off the switch **B** Water lets current pass into the body
   C Water damages the switch's plastic **D** Wet skin makes the switch heat up fast

10. MATHS · CLASS 11
    How many radians is an angle of 90°?

    A π **B** π/2
    C π/3 **D** 2π

11. ENGLISH · BASICS
Asked about the mistake, the cadet gave a candid answer. As used here, "candid" most nearly means

A honest
B short
C rude
D clever

12. SCIENCE · CLASS 11
What is the mass of 0.5 mol of carbon dioxide? (C = 12, O = 16)

A 11 g
B 22 g
C 44 g
D 88 g

13. MATHS · BASICS
Which of these is a perfect square?

A 1,024
B 1,210
C 1,440
D 1,690

14. ENGLISH · BASICS
Choose the correct word: One of my friends ___ a pilot.

A are
B is
C were
D have been

15. SCIENCE · BASICS
An electric kettle draws a current of 5 A from a 230 V supply. What is its power, in watts?

Type-in answer: _____

16. MATHS · CLASS 11
How many two-letter arrangements, with no letter repeated, can be made from the letters A, B, C and D?

Type-in answer: _____

17. ENGLISH · BASICS
Hand is to glove as foot is to ___

A toe
B sock
C walk
D leg

18. SCIENCE · CLASS 11
Which of these elements, all in Group 1, has the largest atomic radius?

A Lithium
B Sodium
C Potassium
D Rubidium

19. MATHS · BASICS
A town's population of 20,000 grows by 10% every year. What will it be after 2 years?

A 24,000
B 24,200
C 24,400
D 26,000

20. ENGLISH · BASICS
"Book the tickets today; the fares go up next week." This advice assumes that

A fares will never come down again
B the train is already nearly full
C the listener still plans to go
D next week is a public holiday

21. SCIENCE · BASICS
On a humid day, drops of water form on the outside of a glass of iced water. This water comes from

A water seeping out through the glass
B the glass giving off moisture
C water vapour in the air condensing
D ice melting inside the glass

22. MATHS · CLASS 11

What is the 6th term of the geometric progression 5, 10, 20, ...?

A 80 **B** 120

C 160 **D** 320

23. ENGLISH · BASICS

In a code, PEN is written as QFO. How is INK written in the same code?

A HMJ **B** KPM

C JOL **D** JNL

24. SCIENCE · CLASS 11

A 1,000 kg car moving at 20 m/s is brought to rest in 5 seconds. What average braking force acts on it?

A 400 N **B** 2,000 N

C 4,000 N **D** 100,000 N

25. MATHS · BASICS

One card is drawn at random from a well-shuffled pack of 52. What is the probability that it is a king or a heart?

A 1/13 **B** 1/4

C 17/52 **D** 4/13

26. ENGLISH · BASICS

Pointing to a young man, Neha said, "His mother is the only daughter of my mother." How is the man related to Neha?

A Brother **B** Son

C Nephew **D** Cousin

27. SCIENCE · BASICS

Lights in a house are wired in parallel, not in series. The main reason is that in parallel

A the house draws a smaller total current **B** each bulb glows dimly and saves energy

C one switch turns every bulb on together **D** each bulb gets full voltage on its own

28. MATHS · CLASS 11

What is the coefficient of x^2 in the expansion of $(1 + x)^5$?

Type-in answer: _____

29. ENGLISH · BASICS

Every pilot in the squadron has passed the night-flying test. Arjun has not passed the night-flying test. Which statement must be true?

A Arjun will pass the test later **B** Arjun is not a pilot in the squadron

C Arjun is a pilot in the squadron **D** No pilot in the squadron flies by day

30. SCIENCE · CLASS 11

A motor lifts a 50 kg load through 10 m in 5 seconds. Taking $g = 10\ m/s^2$, what power does it deliver, in watts?

Type-in answer: _____

Answers

1	**C. 720**	Each number is the one before multiplied by 2, then 3, then 4, then 5. The next step is × 6, and 120 × 6 = 720.
2	**A. opaque**	Transparent means light passes through so you can see through it; opaque means light does not pass through. "Clear" means the same as transparent.
3	**A. the dust stays put as the carpet moves**	The carpet is suddenly pushed away, but the dust tends to stay at rest (inertia), so it is left behind and falls out.
4	**5**	A ∪ B = {2, 4, 6, 8, 10}, which has 5 elements. Adding 3 + 4 = 7 counts 4 and 6 twice. This is a type-in question.
5	**B. courage**	He was praised, so the missing word must be a good quality shown during a rescue. Only courage fits; the rest would be criticised.
6	**40**	Its average speed is (0 + 20) ÷ 2 = 10 m/s, so in 4 s it covers 40 m. Using the final speed for the whole time gives 80 m.
7	**3400**	15% of ₹4,000 is ₹600, so the bicycle sells for 4,000 − 600 = ₹3,400. This is a type-in question.
8	**D. had left**	Two past events are compared, and the train left first. The earlier of two past actions takes the past perfect: "had left".
9	**B. Water lets current pass into the body**	Ordinary water conducts electricity, so moisture can carry current from the switch into the body and cause a shock.
10	**B. π/2**	180° is π radians, so 90° is half of that: π/2. π is a straight angle and 2π a full turn.
11	**A. honest**	Candid means frank and truthful, saying things openly. A candid answer may also be short or blunt, but neither is the meaning of the word.
12	**B. 22 g**	The molar mass of CO_2 is 12 + 2 × 16 = 44 g/mol, so 0.5 mol is 22 g. 44 g is a whole mole; 88 g multiplies by 2 instead of halving.
13	**A. 1,024**	1,024 = 32 × 32. The others fall between squares: 34^2 = 1,156 and 35^2 = 1,225; 37^2 = 1,369 and 38^2 = 1,444; 41^2 = 1,681 and 42^2 = 1,764.
14	**B. is**	The subject is "one", which is singular, so the verb is "is". "Friends" is plural but only describes which group the one belongs to.
15	**1150**	Power is voltage × current: 230 V × 5 A = 1,150 W. This is a type-in question: the answer is typed as a number.
16	**12**	The first letter has 4 choices and the second 3, so $^4P_2$ = 4 × 3 = 12. Counting AB and BA as one gives 6, the number of selections.
17	**B. sock**	A glove is worn on the hand to cover it; a sock is worn on the foot. A toe is part of the foot and walking is what it does.
18	**D. Rubidium**	Atomic radius increases down a group, as each element adds a new shell of electrons. Rubidium is lowest of the four, so it is largest.

19	**B. 24,200**	After one year: 22,000. The second year's 10% is taken on 22,000, adding 2,200, so 24,200. 24,000 forgets the growth compounds.
20	**C. the listener still plans to go**	Advice to book early only makes sense if the listener is going to travel. The other statements are not needed for the advice to hold.
21	**C. water vapour in the air condensing**	The cold glass cools the air touching it, and the water vapour in that air condenses into drops. Glass does not let water through.
22	**C. 160**	The nth term is $ar^{n-1} = 5 \times 2^5 = 160$. 320 uses 2^6, one doubling too many; 80 stops one too early.
23	**C. JOL**	Each letter moves one step forward: P→Q, E→F, N→O. So I→J, N→O, K→L, giving JOL. HMJ moves each letter back instead.
24	**C. 4,000 N**	Force = change in momentum ÷ time = (1000 × 20) ÷ 5 = 4,000 N. 100,000 N multiplies by the time instead of dividing.
25	**D. 4/13**	There are 4 kings and 13 hearts, but the king of hearts is in both, so 16 cards qualify: 16/52 = 4/13. Counting it twice gives 17/52.
26	**B. Son**	The only daughter of Neha's mother is Neha herself. So the man's mother is Neha, and he is her son. "Nephew" assumes a sister.
27	**D. each bulb gets full voltage on its own**	In parallel every bulb gets the full mains voltage and has its own path, so each can be switched on or off without affecting the others.
28	**10**	The term in x^2 is ${}^5C_2\, x^2 = 10x^2$, so the coefficient is 10. ${}^5C_2 = 5 \times 4 \div 2 = 10$. This is a type-in question.
29	**B. Arjun is not a pilot in the squadron**	If Arjun were a pilot in the squadron he would have passed the test. He has not, so he is not one. Nothing is said about the future.
30	**1000**	Work done = mgh = 50 × 10 × 10 = 5,000 J; power = work ÷ time = 5,000 ÷ 5 = 1,000 W. This is a type-in question.